SI Session Planning Form

SI Leader: Brandon Cobb
Session Week: 17
Session#: 17
Instructor: Professor Tramontozzi
Old Plans: SharePoint
Session Goal: Students will review and apply core concepts of concentration and solution chemistry, including calculating molarity, mass percent, volume percent, and mass/volume percent, as well as performing dilution calculations and using solubility rules to determine saturation levels and predict precipitation. They will analyze factors that influence solubility, such as temperature and solute–solvent polarity, and explore colligative properties including freezing point depression and boiling point elevation for electrolytes and nonelectrolytes. Students will strengthen their proficiency in stoichiometry by balancing equations, applying atomic masses, chemical formulas, Avogadro's number, and mole ratios, identifying limiting reactants, and determining theoretical yields. They will classify reaction types—synthesis, decomposition, single and double replacement, combustion, neutralization, and redox—and apply gas laws and kinetic molecular theory to relationships among pressure, volume, temperature, moles, and molecular motion. Students will refine quantitative skills through unit conversions, algebraic manipulation of exact and inexact numbers, correct use of significant figures in multistep calculations, measurement techniques using instruments with fixed uncertainties, proper recording of units, and scientific notation. Additionally, they will deepen their understanding of atomic and subatomic structure, including electron orbitals. By the end of this unit, students will distinguish dissociation from ionization, calculate pH, pOH, ([H_3O^+]), $and([OH^-]), determinepercentionizationofweakacids, interpret(K_a)andpKavalues, identifyconjugateacid–base$

Opener

Take Attendance

Collaborative Learning Techniques	Turn to a Partner
Learning Strategy	Ice Breaker (Two truths and a Lie)
Time	11:30—11:35
Materials	Notes, homework, whiteboard, markers, ready to speak
Lecture/Textbook/Old Plans/Slide References	Class roster

Definitions

a. **Turn to a Partner:** Group members work with a partner on an assignment or discussion topic. This technique works best with group participants who have already been provided with enough background on a subject that they can immediately move to a discussion with their partner without previewing or reviewing concepts.

b. **Ice Breaker:** Each student states two true facts and one false statement related to chemistry concepts (or general academic experiences), and the group guesses which statement is false. This activity encourages peer interaction, sparks discussion, and lightly reviews content in a fun, low-pressure way.

Checking for Understanding (questions))

- What is the primary goal of this activity?
- How do students know if their solution at a station is correct? .
- What should students do if they are unsure about a question?
- How does moving between stations benefit student learning?
- What is the importance of showing work on the team slip?
- How are bonus points awarded during the activity? n.
- What is the role of the facilitator at each station?
- How should teams handle disagreements while solving a question?
- Why is it important that only verified answers are recorded on the team slip?
- What is the wrap-up discussion intended to accomplish?

Activity 1

Collaborative Learning Techniques	Group Discussion
Learning Strategy	Lecture Review
Time	11:40—12:05
Materials	Notes, homework, whiteboard, markers, ready to speak, periodic table
Lecture/Textbook/Old Plans/Slide References	Chapters 1-10

Definitions

a. **Group Discussion:** One person in the group is asked to present on a topic or review material for the group and then lead the discussion for the group. This person should not be the regular group leader.

b. **Lecture Review:** As a group summarize the lecture from the previous class. You may have to provide prompts for the students. For example, "The first concept discussed was Civil Liberties and Public Policy, what did the profressor highlight regarding this?" You may want to ask them to try summarizing without looking at their notes; however, if they are having a difficult time remembering, tell them to refer to their notes.

Activity Overview

a. **Purpose:** Provide students with a collaborative, lecture-integrated review session that strengthens comprehension, encourages active participation, and reinforces key concepts from the most recent lesson. Students work together to analyze prompts, discuss big ideas, and apply them to example scenarios.

b. **Structure:**

- Students work in small groups (3–5).
- The review proceeds in several timed "rounds," each focused on a major idea from the lecture (e.g., mechanisms, definitions, graphical reasoning, model interpretation).
- Each round involves a shared prompt projected for the class (or distributed as slips), followed by structured group discussion and a brief written response.
- After each round, the instructor leads a whole-class debrief before transitioning to the next concept.

c. **Materials:**

- Prepared question prompts for each review round.
- Group response sheets or whiteboards.
- Timer or projected countdown.
- Optional visual supports (figures, graphs, chemical structures).

Instructions (detailed activity breakdown below) continued...

Facilitation Instructions

a. **Group Management:**

- Assign students to balanced groups based on seating or prior performance.
- Encourage each group to designate rotating roles (scribe, presenter, checker, time-keeper).

b. **Round Interaction:**

- Present the prompt for the round and start the timed interval.
- Groups discuss the question, synthesize reasoning, and record a short, unified response.
- Encourage equal participation within groups; visit groups and ask probing questions.
- When time is up, instruct groups to finalize responses before the debrief.

c. **Instructor Role:**

- Circulate to observe discussions and identify misconceptions.
- Use targeted guiding questions to stimulate reasoning.
- Collect insights during group work to address during whole-class debriefs.
- Maintain pacing across rounds to keep the review session energetic and focused.

Rules and Expectations

a. All group members must contribute meaningfully to each round.

b. Responses must reflect consensus, not individual work.

c. Respectful discourse is expected; challenges to ideas should be evidence-based.

d. Groups should stay focused on the given prompt during each timed interval.

e. Written responses should be concise, clear, and connected to lecture concepts.

Instructions (detailed activity breakdown below) continued...

Scoring and Feedback

a. Groups earn points for each round based on accuracy, completeness, and explanation quality.

b. Bonus points may be awarded for:

- Exceptional clarity or creative analogies.
- Effective visual representations (diagrams, mechanisms, models).
- Strong group participation observed by the instructor.

c. Post-round debriefs provide immediate correction and conceptual reinforcement.

d. Optionally maintain a leaderboard to encourage engagement (without high-stakes pressure).

Wrap-Up and Reflection

a. After all rounds conclude, collect group responses to analyze trends in understanding.

b. Highlight areas where multiple groups struggled and re-explain key ideas.

c. Lead a brief reflection discussion addressing:

- Which lecture topics felt most or least comfortable.
- How group discussion deepened understanding of the material.
- What strategies students plan to use for individual study.

d. Conclude by connecting the reviewed concepts to upcoming assessments or applications.

Content (fully worked out problems & answers)

(1) For the reaction A + heat → B, which statement is correct?

a. Adding heat increases [B]

b. Adding heat decreases [B]

c. Removing heat increases [B]

d. Heat has no effect on equilibrium

(2) Which of the following best describes a saturated solution at equilibrium?

a. Solute is dissolving faster than it crystallizes

b. Solute and solvent are reacting chemically

c. Dissolving and crystallization occur at equal rates

d. No solute particles are moving

(3) Calcium hydroxide reacts with hydrochloric acid to form calcium chloride and water. Which statement best describes the acid-base behavior occurring in this reaction?

$Ca(OH)_2 + 2HCl \rightarrow CaCl_2 + 2H_2O$

a. HCl donates protons to OH^- ions, forming water as Ca^{2+} pairs with Cl^-

b. $Ca(OH)_2$ donates protons to HCl, forming hydrogen gas

c. Both reactants accept protons, preventing water formation

d. Ca^{2+} acts as the proton donor while Cl^- acts as the proton acceptor

(4) When comparing HF and HCl, HF has a much higher boiling point. Which force best explains this difference?

a. Dipole-dipole interactions

b. Hydrogen bonding

c. London dispersion forces

d. Covalent bonding

(5) Which of the following increases when temperature increases for a gas at constant volume?

a. Pressure

b. Molecular mass

c. Intermolecular forces

d. Volume

(6) A chemist has 50.0 mL of 2.0 M HCl solution. She adds water to dilute it to a final volume of 200.0 mL. What is the molarity of the diluted solution?

a. 0.25 M

b. 0.50 M

c. 1.0 M

d. 2.0 M

(7) An unsaturated solution contains less solute than the maximum that can dissolve at that temperature.

True **False**

(8) What happens to an atom's oxidation number when it is oxidized?

a. It increases

b. It decreases

c. It stays the same

d. It becomes zero

(9) Which of the following solutes is generally soluble in water according to solubility rules?

a. $BaSO_4$

b. AgCl

c. KNO_3

d. PbI_2

(10) If $Q > K_{eq}$ for the reaction F $\rightleftharpoons$ G, which direction will the system shift?

a. Toward reactants

b. Toward products

c. No shift

d. The reaction stops

(11) 0.250 mol of oxygen gas is collected at 740. mmHg and 18.0°C in a 2.00 L flask. What would be the pressure at 0.500 mol in same flask and same T? (assume ideal)

a. 1480 mmHg

b. 740. mmHg

c. 370. mmHg

d. 1110 mmHg

(12) A Brönsted-Lowry acid is defined as a substance that:

a. Donates a proton

b. Accepts a proton

c. Releases electrons

d. Accepts electrons

(13) A buffer solution is prepared using 0.120 mol NH_3 and 0.080 mol NH_4^+ with $K_a = 5.6 \times 10^{-10}$ for NH_4^+. After calculating the pH, what is the corresponding $[OH^-]$?[0.2cm] [0.2cm]

a. pH = 9.32, $[OH^-] = 2.13 \times 10^{-5}$ M

b. pH = 9.43, $[OH^-] = 1.10 \times 10^{-5}$ M

c. pH = 9.43, $[OH^-] = 2.68 \times 10^{-5}$ M

d. pH = 8.90, $[OH^-] = 7.99 \times 10^{-6}$ M

(14) Which of the following would double the (m/v)% of a solution?

a. Double solute, same volume

b. Double volume, same solute

c. Halve solute, double volume

d. Keep solute and volume constant

(15) A 5% (m/v) glucose solution contains 5 g of glucose in:

a. 5 mL of water

b. 100 mL of water

c. 100 mL of solution

d. 95 mL of solution

(16) The density of aluminum is 2.70 g/cm^3. What is the mass of a block of aluminum with a volume of 15.0 cm^3?

a. 40.5 g

b. 5.56 g

c. 17.7 g

d. 2.70 g

(17) Convert 0.750 L into cubic centimeters. (Use 1 L = 1000 cm^3)

a. 75.0 cm^3

b. 750. cm^3

c. 7.50 cm^3

d. 0.00750 cm^3

(18) Identify the reaction type:

$2H_2O_2 \rightarrow 2H_2O + O_2$

a. Combination

b. Combustion

c. Decomposition

d. Exchange

(19) What term describes the process by which a solid dissolves until no more can dissolve under given conditions?

a. Crystallization

b. Saturation

c. Dilution

d. Precipitation

(20) A cylinder contains 0.850 mol of gas at 2.50 atm. What is the temperature (K) if the volume is 15.0 L?

a. 210. K

b. 298 K

c. 425 K

d. 537 K

(21) A student notices that hexane and water do not mix when shaken together. Which type of intermolecular forces explains this behavior?

a. London dispersion forces dominate in hexane, but water is polar.

b. Both substances form hydrogen bonds with each other.

c. Hexane is polar, and water is nonpolar.

d. Dipole-dipole interactions attract the two liquids together.

(22) Which of the following represents a redox reaction?

a. $HCl + NaOH \rightarrow NaCl + H_2O$

b. $Zn + 2HCl \rightarrow ZnCl_2 + H_2$

c. $CaCO_3 \rightarrow CaO + CO_2$

d. $NaOH + HNO_3 \rightarrow NaNO_3 + H_2O$

(23) A solution has $[H_3O^+] = 2.22 \times 10^{-6}$ M. Calculate the pH, pOH, and $[OH^-]$.

(24) A buffer is prepared using 0.250 M HCO_3^- and 0.400 M CO_3^{2-} ($pK_a2 = 10.33$). What are the buffer pH and corresponding pOH?

a. pH = 10.13, pOH = 3.87

b. pH = 10.33, pOH = 3.67

c. pH = 10.73, pOH = 3.27

d. pH = 10.15, pOH = 3.85

(25) A 425 mL cylinder contains nitrogen at 2.50×10^2 kPa and 22.0°C. What pressure (in atm) will result if the gas is moved to a 1.00 L container and heated to 125°C, assuming moles constant?

a. 0.564 atm

b. 1.41 atm

c. 0.282 atm

d. 2.24 atm

(26) What type of reaction occurs when potassium iodide reacts with lead(II) nitrate to form lead(II) iodide and potassium nitrate?

$2KI + Pb(NO_3)_2 \rightarrow 2KNO_3 + PbI_2$

a. Exchange

b. Combination

c. Decomposition

d. Combustion

(27) When salt dissolves in water, why is this considered a physical change?

a. The process creates a new chemical substance with different bonds.

b. The salt changes state but its chemical composition remains the same.

c. The salt reacts with water to form a new compound.

d. The ions rearrange permanently to form a new molecule.

(28) A buffer is made by mixing 0.200 M HF and 0.350 M NaF ($pK_a = 3.17$). After preparing the buffer, what are the pH and resulting $[H_3O^+]$?

a. pH = 2.93, $[H_3O^+] = 1.18 \times 10^{-3}$ M

b. pH = 3.33, $[H_3O^+] = 4.7 \times 10^{-4}$ M

c. pH = 3.17, $[H_3O^+] = 6.8 \times 10^{-4}$ M

d. pH = 3.60, $[H_3O^+] = 2.5 \times 10^{-4}$ M

(29) A solution has $[H_3O^+] = 3.20 \times 10^{-6}$ M. Its pH is closest to:

a. 4.49

b. 5.49

c. 6.49

d. 7.49

(30) A weak acid has Ka = 3.00×10^{-7}. Which pKa is closest?

a. 3.96

b. 4.22

c. 5.44

d. 6.80

(31) Two rigid containers are connected by a closed valve. One container holds helium at 2.0 atm in 4.0 L, and the other holds neon at 4.0 atm in 2.0 L. When the valve is opened and gases mix, what is the total pressure in the combined 6.0 L system (assume constant temperature)?

a. 2.0 atm

b. 2.7 atm

c. 4.0 atm

d. 6.0 atm

(32) A solution with pH = 2.00 has a hydronium concentration that is how many times larger than a solution with pH = 4.00?

a. 2x

b. 10x

c. 100x

d. 1000x

(33) Which of the following reactions is a redox reaction?

a. $Zn + CuSO_4 \rightarrow ZnSO_4 + Cu$

b. $AgNO_3 + NaCl \rightarrow AgCl + NaNO_3$

c. $CaCO_3 \rightarrow CaO + CO_2$

d. $H_2O + SO_3 \rightarrow H_2SO_4$

(34) Which factor determines how much the freezing point of a solution is lowered?

a. The number of solute particles

b. The solute's color

c. The solvent's taste

d. The solute's shape

(35) What happens to the molarity of a solution when it is diluted with more solvent?

a. It increases

b. It decreases

c. It stays the same

d. It doubles

(36) What is the molarity of a solution containing 5.00 mol of KBr in 2.00 L of solution?

a. 2.00 M

b. 2.50 M

c. 5.00 M

d. 10.0 M

(37) What is the percent yield if 10.0 g of product are expected but only 8.20 g are obtained?

a. 82.0%

b. 92.0%

c. 80.0%

d. 88.0%

(38) Which species is the conjugate acid of NH_3?

a. NH_2^-

b. NH_4^+

c. N^{3-}

d. NH_2OH

(39) Osmosis is the movement of solvent molecules from:

a. Low solute concentration to high solute concentration

b. High solute concentration to low solute concentration

c. High pressure to low pressure

d. High temperature to low temperature

(40) A balloon is filled with a gas mixture containing 0.6 mol O_2 and 0.4 mol N_2 at a total pressure of 1.0 atm. What is the partial pressure of O_2?

a. 0.4 atm

b. 0.6 atm

c. 1.0 atm

d. 1.5 atm

(41) How many grams of KNO_3 are required to prepare 250.0 mL of 0.200 M KNO_3 solution?

a. 2.53 g

b. 5.06 g

c. 10.1 g

d. 20.2 g

(42) Convert 250.0 cm/s to mi/h. (Use 1 mi = 1.609 km)

a. 5.59 mi/h

b. 15.6 mi/h

c. 0.155 mi/h

d. 1.56 mi/h

(43) For the reaction $R + S \rightleftharpoons RS$, which change could temporarily increase Q?

a. Adding RS

b. Removing R

c. Removing S

d. Diluting all species equally

(44) In the reaction $HCl + H_2O \rightarrow H_3O^+ + Cl^-$, which is the Brönsted-Lowry acid?

a. HCl

b. H_2O

c. H_3O^+

d. Cl^-

(45) Which of the following changes how much a solution's freezing point is lowered?

a. How much solute is added

b. The type of solute (number of particles it makes)

c. A & B

d. The color of the solute

(46) Convert 745 torr to atm. (Use 1 atm = 760 torr)

a. 0.980 atm

b. 1.02 atm

c. 0.760 atm

d. 0.745 atm

(47) A 25% (v/v) ethanol solution means:

a. 25 mL ethanol in 100 mL solution

b. 25 mL ethanol in 100 mL solvent

c. 25 g ethanol in 100 mL solution

d. 25 mL solution in 100 mL ethanol

(48) Which species is amphiprotic?

a. OH^-

b. HSO_4^-

c. Cl^-

d. NH_4^+

(49) A buffer is prepared by mixing 0.300 mol acetic acid ($pK_a = 4.76$) with 0.450 mol sodium acetate in 1.00 L solution. What is the pH of the buffer, and what is the corresponding pOH?

a. pH = 4.94, pOH = 9.06

b. pH = 4.76, pOH = 9.24

c. pH = 4.58, pOH = 9.42

d. pH = 5.12, pOH = 8.88

(50) For the equilibrium $PCl_5(g) \rightleftharpoons PCl_3(g) + Cl_2(g)$, increasing pressure causes the reaction to shift:

a. Left, toward PCl_5

b. Right, toward PCl_3 and Cl_2

c. No shift because gases are unaffected by pressure

d. Toward whichever side has the higher total mass

(51) For each of the following, select the option that correctly identifies each property as physical (P) or chemical (C): [0.2cm]

a. melting point of ice

b. flammability of hydrogen gas

c. density of aluminum

d. ability of iron to rust

Choices: A) P, C, P, C B) C, P, C, P C) P, P, C, C D) C, C, P, P

(52) For each of the following, select the option that correctly classifies each as a pure substance (PS) or a mixture (M): [0.2cm]

a. sodium chloride

b. air

c. copper

d. seawater

Choices: A) PS, M, PS, M B) M, PS, M, PS C) PS, PS, M, M D) M, M, PS, PS

(53) A gas occupies 500. mL at 25°C. What volume will it occupy at 100.°C if the pressure remains constant?

a. 583 mL

b. 625 mL

c. 665 mL

d. 740. mL

(54) Convert 5850 cm^2 to m^2.

a. 0.0585 m^2

b. 5.85 m^2

c. 585 m^2

d. 0.000585 m^2

(55) Which statement correctly defines ionization for a base in water?

a. Forming OH^- by reacting chemically with water rather than releasing pre-existing ions

b. Separating into pre-existing ions without chemical change

c. Producing H_3O^+ by transferring a proton to water

d. Splitting into neutral molecules without forming ions

(56) Which term describes an acid that ionizes partially in water?

a. Strong acid

b. Weak acid

c. Strong base

d. Neutral compound

(57) Osmotic pressure depends on which of the following factors?

a. The number of solute particles in solution

b. The color of the solute

c. The shape of the solvent molecules

d. The size of the container

(58) A buffer is formed by mixing 0.500 mol formic acid ($pK_a = 3.75$) with 0.300 mol sodium formate. What are the buffer pH and the resulting $[H_3O^+]$?

a. pH = 3.52, $[H_3O^+] = 3.0 \times 10^{-4}$ M

b. pH = 3.75, $[H_3O^+] = 1.8 \times 10^{-4}$ M

c. pH = 3.32, $[H_3O^+] = 4.8 \times 10^{-4}$ M

d. pH = 3.90, $[H_3O^+] = 1.3 \times 10^{-4}$ M

(59) A tectonic plate moves 860 km per day. What is its speed in m/s?

a. 9.95 m/s

b. 0.0995 m/s

c. 9.95×10^3 m/s

d. 2.47 m/s

(60) Which salt will produce a basic solution upon dissolution due to its parent species?

a. NaF

b. NH_4Br

c. KCl

d. $Ca(NO_3)_2$

Checking for Understanding (questions & answers)

- How can molarity be used to determine the number of moles of solute present in a given volume of solution?
- What steps are required to calculate mass percent, volume percent, or mass/volume percent from experimental data?
- How do dilution calculations relate initial and final concentrations and volumes of a solution?
- What evidence indicates that a solution is saturated, unsaturated, or supersaturated at a particular temperature?
- How do solubility rules help predict whether a precipitate will form during a double replacement reaction?
- Why does increasing temperature typically increase the solubility of most solid solutes but decrease the solubility of gases?
- How does solute–solvent polarity affect dissolution and miscibility?
- How does adding a nonvolatile solute affect the freezing point of a solvent?
- Why does dissolving an ionic compound typically lower the freezing point more than dissolving a covalent compound?
- What factors determine the magnitude of freezing point depression in a solution?

- Why does the identity of a solute, not just its amount, influence colligative properties?
- How does the boiling point of a solution compare to the boiling point of its pure solvent?
- Why does adding solute increase the boiling point of a liquid?
- Which physical property remains unchanged when the freezing point of a solution is lowered?
- Why does temperature itself not change the amount of freezing point depression?
- Why do electrolytes and nonelectrolytes produce different magnitudes of colligative effects?
- How can a balanced chemical equation be used to determine mole ratios for stoichiometric calculations?
- Why is balancing chemical equations required to satisfy conservation of mass?
- How can stoichiometric relationships be used to determine theoretical yield?
- How can you identify the limiting reactant when given initial quantities of multiple reactants?
- Why do ionic compounds dissociate into ions in water, while many molecular compounds do not?
- How does solubility influence whether certain ions in a double replacement reaction form a solid precipitate?
- Why is water considered an effective solvent for ionic and polar solutes?
- How does molecular polarity explain why nonpolar substances like oil do not mix with water?
- Why do stronger intermolecular forces correspond to higher boiling points?

- How can temperature changes be used to determine whether a process is endothermic or exothermic?
- Why is combustion classified as an exothermic oxidation reaction?
- What evidence suggests a chemical change has occurred rather than a physical change?
- Why is neutralization considered a double replacement reaction, and how does it differ from precipitation and redox reactions?
- How can synthesis, decomposition, single replacement, and double replacement reactions be distinguished based on reactants and products?
- How does the activity series help predict whether a single replacement reaction will occur?
- How do gas laws relate pressure, volume, temperature, and number of moles for a gas sample?
- How does the kinetic molecular theory explain changes in gas pressure and temperature?

- Why must unit conversions and significant-figure rules be followed carefully in multistep calculations?
- How do fixed instrumental uncertainties influence how measurements should be recorded?
- How can Avogadro's number be applied to convert between moles and number of particles?
- How do electron orbitals describe the probable locations of electrons within atoms?
- What is the difference between dissociation and ionization in aqueous solutions?
- How can pH, pOH, $[H_3O^+]$, and $[OH^-]$ be calculated from one another?
- How can percent ionization be determined for a weak acid given its concentration and K_a?
- How do K_a and pK_a values indicate the strength of an acid?
- How do conjugate acid–base pairs differ within a Brønsted–Lowry framework?
- How do polyprotic acids undergo multiple ionization steps, and how does this affect solution pH?
- How can salt hydrolysis be used to predict whether a solution formed from a dissolved salt will be acidic, basic, or neutral?
- How can consistency checks and correct units ensure that quantitative solution calculations are valid?

Activity 2

Collaborative Learning Techniques	Jigsaw
Learning Strategy	Predict Test Questions
Time	12:05—12:30
Materials	Pencil, eraser, calculator, periodic table
Lecture/Textbook/Old Plans/Slide References	Chapter 10: Acid and Bases

Definition

a. **Jigsaw:** Jigsaws, when used properly, make the group as a whole dependent upon all of them in subgroups. Each group provides a peice of the puzzle. Group members are broken into smaller groups. Each small group works on some aspect of the same problem, question, or issue. They then share their part of the puzzle with the large group.

b. **Predict Test Questions:** Put students in groups of 2 or 3 and assign them to write a test question for a specific topic, ensuring that all topics have been convered. Ask students to write their questions on the board or on an overhead for discussion (would the professor ask the question? What is the answer? etc.). Students will have the benefit of learning to think like the teacher and they'll be able to see additional questions that other students have writen.

Instructions (detailed activity breakdown below)

a. **Setup:**

- Divide students into small groups and assign each group a role structure (e.g., scribe, presenter, checker, timekeeper).
- Provide each group with a blank question card for writing their original test question.
- Give each student an individual worksheet to work on while the instructor solves the group-generated questions.

b. **Formulating the Question (5 minutes):**

- Each group collaborates to write one test question related to current unit topics (calculations, concepts, reaction types, etc.).
- The question must include:
  - A clearly stated prompt.
  - Any necessary data or diagrams.
  - A final answer key written *lightly* on the back for instructor use only.
- Groups must agree unanimously that the question is fair, solvable, and aligned with recent course content.
- When time is up, the instructor collects all question cards and begins solving them.

c. **Independent Work Period (Instructor Solves Questions):**

- While the instructor works through each group's question, students complete their individual worksheets quietly.
- Groups may discuss the worksheet only after all members finish the same section.
- The instructor prepares corrected solutions and confirms the correct answer for each group-generated test question.

1. **Question Exchange and Group Challenge:**
   - Once solved, the instructor returns each group *another group's* test question.
   - Groups must work collaboratively using assigned roles to solve the received question.
   - All members must agree on the final answer before reporting it.
   - Each group provides its final answer verbally to the instructor when ready.
2. **Scoring:**
   - A group earns **1 point** for submitting the correct answer to the exchanged question.
   - A group earns **1 bonus point** if their originally written test question is solved correctly by the receiving group.
   - Groups with full agreement and strong justification may receive optional teamwork bonus credit.
   - The team with the highest point total wins the challenge.
3. **Rules:**
   - Groups must produce their question collaboratively; no individual writing without group input.
   - All group members must agree on submitted answers—consensus is required.
   - Worksheet work during the instructor-solving phase must be independent.
   - Direct sharing of answers between groups is prohibited.
   - Questions must be original and may not copy textbook or online sources verbatim.
4. **Wrap-Up:**
   - Review a selection of the student-generated questions as a whole class.
   - Highlight creative question design, common misconceptions, and strong reasoning.
   - Discuss how generating questions reinforced deeper understanding of unit concepts.
   - Encourage reflection on teamwork, consensus-building, and accuracy in problem solving.

Content (fully worked out problems & answers)

1. **Chapter 1 – Basic Concepts About Matter**
   - Classification of matter: element, compound, mixture (simple flow diagram).
   - States of matter and particle spacing.
   - Physical vs. chemical properties and changes (icon examples).
   - Density formula and basic interpretation (mass/volume block sketch).
2. **Chapter 2 – Measurements in Chemistry**
   - Metric units used in chemistry (g, L, m) and common prefixes (k, c, m).
   - Significant figure rules (quick chart: zeros counting vs. not).
   - Dimensional analysis (one-step unit conversion map).
   - Temperature scales: Celsius and Kelvin (short conversion diagram).
3. **Chapter 3 – Atomic Structure and the Periodic Table**
   - Structure of the atom: protons, neutrons, electrons (basic Bohr-style diagram).
   - Atomic number, mass number, and isotopes (notation with symbols).
   - Organization of periodic table: groups, periods, metals/nonmetals.
   - Very basic periodic patterns: valence electrons by group, simple charge patterns for ions.
4. **Chapter 4 – Ionic Bonding**
   - How ions form (metal→cation, nonmetal→anion).
   - Simple electron transfer diagrams.
   - Writing ionic formulas using charge balance.
   - Naming ionic compounds (metal + nonmetal ending in -ide).
   - Recognizing common polyatomic ions (small table only).
5. **Chapter 5 – Covalent Bonding**
   - Covalent bonds as shared electrons (simple dot diagrams).
   - Single vs. double bonds (basic Lewis examples).
   - Electronegativity and polar covalent bonds (arrow-with- symbols).
   - VSEPR shapes: linear, bent, tetrahedral
   - Naming simple molecular compounds (prefix system).

1. **Chapter 6 – Chemical Calculations**
   - Formula mass / molar mass (periodic table extraction).
   - The mole and Avogadro's number (conceptual diagram with "counting unit").
   - Mole conversions (grams moles particles).
   - Balanced equations and mole ratios (visual ratio arrows).
   - Basic limiting reactant setups (simple table comparing amounts).
2. **Chapter 7 – Gases, Liquids, and Solids**
   - Particle diagrams for each state of matter.
   - Gas variables (P, V, T, n) and direct/inverse relationships.
   - Basic gas laws: Boyle's, Charles', Combined Gas Law (no derivations).
   - Intermolecular forces: dispersion, dipole, H-bonding (simple ranking diagram).
   - Heating curve sketch (solid→liquid→gas steps).
3. **Chapter 8 – Solutions**
   - Solute vs. solvent (particle picture).
   - Types of solutions: electrolyte vs. nonelectrolyte (simple ion break-apart diagram).
   - Concentrations: molarity, mass%, volume%, mass/volume%.
   - Dilutions (MV = MV).
   - Solubility and saturation (curve showing solubility vs. temperature).
   - Colligative properties at an intro level: FPD and BPE (directional arrows only).
4. **Chapter 9 – Chemical Reactions**
   - Evidence of a reaction (gas, color change, precipitate, energy change).
   - Reaction types: synthesis, decomposition, single replacement, double replacement, combustion.
   - Predicting products using simple patterns (A + B → AB, AB → A + B, etc.).
   - Precipitation diagram (two aqueous ions forming a solid).
   - Energy diagrams: exothermic vs. endothermic (simple up/down curve).
5. **Chapter 10 – Acids, Bases, and Salts**
   - Properties of acids and bases (taste/feel cautions + conductivity).
   - Arrhenius definitions (H producers, OH producers).
   - Very basic Brønsted concept (acid donates H).
   - pH scale (0–14 color bar).
   - Neutralization reaction (acid + base → salt + water).
   - Strong vs. weak acids (simple dissociation comparison).
   - Salt solutions: acidic, basic, neutral (simple prediction table).

Checking for Understanding (questions)

a. How do you distinguish between elements, compounds, and mixtures using particle-level representations?

b. How can you determine whether a change is physical or chemical based on observable evidence?

c. How do you calculate density from mass and volume, and how can density be used to identify an unknown substance?

d. How do metric prefixes relate to base units, and how do you convert between units using factor–label (dimensional analysis) methods?

e. How do you determine the number of significant figures in a measurement, and how do rounding rules differ for multiplication/division versus addition/subtraction?

f. How do you correctly record the value on a graduated cylinder, thermometer, or balance, including the estimated digit?

g. How does the atomic number determine an element's identity, and how do isotopes differ in atomic mass but not chemical behavior?

h. How do you calculate the number of protons, neutrons, and electrons in neutral atoms and simple ions?

i. How does the arrangement of the periodic table help predict the charges of ions formed by main-group elements?

j. How do metals and nonmetals form ionic compounds through electron transfer, and how do you write formulas that balance charges?

k. How do you draw simple Lewis dot structures for main-group elements and use them to predict bonding patterns?

l. How do differences in electronegativity determine whether bonds are nonpolar covalent, polar covalent, or ionic?

m. How do molecular shapes such as linear, bent, trigonal pyramidal, and tetrahedral influence molecular polarity?

n. How do you calculate molar mass from a chemical formula, and how do you use it to convert between grams, moles, and number of particles?

o. How do you balance a chemical equation, and why is balancing necessary to satisfy the law of conservation of mass?

p. How do you use mole ratios from a balanced equation to relate quantities of reactants and products?

q. How do you determine the limiting reactant by comparing available moles of each reactant, and how do you use it to calculate theoretical yield?

1. How do particle diagrams illustrate the differences between gases, liquids, and solids in terms of spacing and motion?
2. How do gas laws such as Boyle's and Charles's describe relationships between P, V, and T, and how do you choose the correct law for a given situation?
3. How does increased kinetic energy at higher temperatures influence gas pressure, diffusion, and average molecular speed?
4. How do you distinguish electrolytes from nonelectrolytes by their behavior in water, and how do ion diagrams represent dissociation?
5. How do you calculate concentration using molarity, mass percent, volume percent, and mass/volume percent for different types of solutions?
6. How do you determine whether a solution is unsaturated, saturated, or supersaturated using a solubility curve?
7. How do you apply solubility rules to predict whether a precipitate forms when two aqueous solutions are mixed?
8. How do you compute dilution values using $M_1V_1 = M_2V_2$, and what quantity remains constant during a dilution process?
9. How does the number of dissolved particles affect boiling point elevation and freezing point depression, and why do ionic compounds cause larger changes?
10. How can you classify reactions as synthesis, decomposition, single replacement, double replacement, or combustion based on their reactants and products?
11. How do you identify whether energy is absorbed or released by examining simple reaction diagrams for endothermic and exothermic changes?
12. How do Arrhenius acids and bases differ, and how does the formation of hydronium allow you to represent acids in aqueous solution?
13. How do you determine pH from hydronium concentration, and how do you convert between pH, pOH, $[H_3O^+]$, and $[OH^-]$ using K_w?
14. How do strong acids and strong bases behave differently from weak ones in terms of dissociation and resulting pH?
15. How do you identify conjugate acid–base pairs in a Brønsted–Lowry reaction, and how do proton transfers define these relationships?
16. How do diprotic and triprotic acids ionize in steps, and why does each successive step release fewer hydronium ions?
17. How do salts formed from strong or weak acids and bases influence the pH of a solution through hydrolysis?
18. How do you verify that calculated pH, pOH, $[H_3O^+]$, and $[OH^-]$ are internally consistent and properly labeled with units or unitless conventions?

Closer

Collaborative Learning Techniques	Group Survey
Learning Strategy	Brain dump
Time	12:30—12:32
Materials	Index cards, writing utensils
Lecture/Textbook/Old Plans/Slide References	Chapters 1-10

Definitions

a. **Group survey:** Each group member is surveyed to discover their position on an issue, problem or topic. This process ensures that each member of the group is allowed to offer or state their point of view.

b. **Brain dump:** This is a great closer for students to sum up their knowledge on the topic(s) from the session. Have the students take out a blank piece of paper /Google Doc and write/type("dump") everything they know relevant from the discussion that day. The SI Leader can look at their work to assess the students' understanding of the course material and have them compare it to each other. It is important to at least discuss with the students so the SI Leader knows the students' comfort levels with the material.

Instructions (please provide a DETAILED activity breakdown below (and/or attached

[itemsep=0.3em]Students are put into small groups and are asked to write down their confidence on a single notecard without names. The cards are evidence of the group's confidence in the content.

Content (fully worked out problems & answers)

- No new worked problems; brain dump.

Checking for Understanding (questions & answers)

- Review confidence levels
- Interpret the variety of confidence and their distribution in groups.
- Get feedback for what they like and don't like
- Check in if the time is suitable for the students

Plan Checklist

- ☐ Variety of Collaborative Learning Techniques and Learning Strategies
- ☐ Chapter/slide# where information is coming from
- ☐ Included timestamps (and buffer time as needed)
- ☐ Details on how leader will break up students
- ☐ Checking for Understanding questions
- ☐ Fully worked out problems and examples for each activity
- ☐ Necessary links required for online implementation (if applicable)
- ☐ Source material correctly cited (if applicable)
- ☐ Plan is uploaded to Teams